

A Deeper Dive in the Gordon–Bogle Return Decomposition

Growth, Discount Rates, and Parameter Uncertainty

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August 20, 2026

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1 Discounted Cash Flow Models

Discounted-cash-flow valuation links the price of an asset to three objects: the cash flows generated by the asset, the rate at which those cash flows are expected to grow, and the

rate at which investors discount them. The Gordon–Shapiro model (Gordon and Shapiro, 1956) compresses these ingredients into a particularly transparent formula. That formula is useful not only as a valuation rule, but also as a bridge from valuation levels to returns: prices change because fundamentals change and because investors change the multiple they are willing to pay for those fundamentals. The latter observation is the basis of the return decomposition popularized by Bogle (1991).

We proceed in six steps. First, we derive the deterministic Gordon formula and obtain Bogle’s cash-flow yield, fundamental growth, and multiple-change components from the exact one-period total-return identity. Second, we allow uncertainty about long-run growth and required returns and derive a conditional multiple governed by the mean discount–growth spread, the relative dispersion of that spread, and its covariance with growth; this isolates the associated Jensen effect on valuation. Third, we add conditionally independent lognormal fluctuations around persistent latent growth and discount-rate centers; this preserves a closed-form geometric-series valuation and isolates a small-volatility wedge. Fourth, we express changes in the multiple in both finite and local form and combine them with realized earnings news to attribute excess returns to conditional carry, the direct earnings surprise, and revisions to long-run growth, required returns, and uncertainty. Fifth, we give an operational multifactor rule for choosing the stock-level Gordon rate m_i , distinguishing current one-period inputs, normalized long-run inputs, and a horizon-specific discount-rate path. Finally, we calibrate the resulting sensitivities for a representative mature stock and compare their magnitudes with firm-level evidence. Throughout, we separate exact identities from local approximations and treat the components as correlated economic attributions, not as orthogonal variance shares.

1.1 Gordon Model

Let

- P_t denote the ex-distribution price of the asset at date t ;
- e_t denote earnings per share and $d_t := \kappa e_t$ the cash flow distributed to shareholders, for a constant payout ratio $\kappa > 0$;
- g denote the constant one-period earnings growth rate;
- m denote the constant one-period required return, or discount rate;
- r_t^f denote the one-period risk-free rate known at date t ;
- $R_{t+1} := (P_{t+1} + d_{t+1})/P_t$ denote the gross return, and $r_{t+1} := R_{t+1} - 1$ the corresponding simple return; and
- $\Delta z_t := z_{t+1} - z_t$ denote the one-period change in any dated quantity z_t .

The constant-payout assumption is a deliberate simplification. In an empirical DCF model one would model payout explicitly or replace earnings with the appropriate free

cash flow. Here the assumption isolates the relation among growth, discounting, and the price-earnings multiple.

Suppose $e_t > 0$ and earnings grow forever at rate $g > -1$,

$$e_{t+s} = e_t(1 + g)^s, \quad s \geq 1, \quad (1.1)$$

and suppose that every future cash flow is discounted at the constant required return $m > -1$. Provided $m > g$, the ratio $(1 + g)/(1 + m)$ lies between zero and one, and the following geometric series converges:

$$P_t = \sum_{s=1}^{\infty} \frac{d_{t+s}}{(1 + m)^s} \quad (1.2)$$

$$= \kappa e_t \sum_{s=1}^{\infty} \left(\frac{1 + g}{1 + m} \right)^s \quad (1.3)$$

$$= \kappa e_t \frac{(1 + g)/(1 + m)}{1 - (1 + g)/(1 + m)} \quad (1.4)$$

$$= \kappa e_t \frac{1 + g}{m - g}. \quad (1.5)$$

Consequently, the Gordon formula written as an earnings multiple is

$$q_t := \frac{P_t}{e_t} = \kappa \frac{1 + g}{m - g}. \quad (1.6)$$

The formula makes the central trade-off transparent. Faster expected growth raises the multiple, whereas a higher required return lowers it. In particular, valuation is very sensitive to either rate when the discount-growth spread $m - g$ is small. When m and g are small in absolute terms, $1 + g \approx 1$, so the convenient low-rate approximation is

$$q_t \approx \frac{\kappa}{m - g}. \quad (1.7)$$

Because the payout ratio is constant, it drops out of changes in $\log q_t$. It nevertheless remains economically relevant through the cash-flow yield, so we retain κ below.

The discount rate m is not merely an interest rate. It is the return that investors require for holding the asset, often called the cost of equity. It therefore contains both compensation for the time value of money and a premium for bearing risk:

$$m = r^f + \text{required risk premium}. \quad (1.8)$$

The deterministic Gordon model treats this required return as constant. In [Section 1.2](#) we make it uncertain, allow its conditional estimate to vary over time, and relate that estimate to standard factor models.

1.1.1 From valuation to returns: Bogle's decomposition

The identity $P_t = q_t e_t$ turns the valuation formula into a return decomposition. Define realized earnings growth by

$$g_{t+1}^{\text{real}} := \frac{e_{t+1} - e_t}{e_t}. \quad (1.9)$$

Also define the realized cash-flow yield over the holding period by

$$y_{t+1} := \frac{d_{t+1}}{P_t} = \frac{\kappa(1 + g_{t+1}^{\text{real}})}{q_t}. \quad (1.10)$$

Because P_{t+1} is the ex-distribution price, the investor receives both the ending price and the distribution. The gross return is therefore, universally,

$$R_{t+1} = \frac{P_{t+1} + d_{t+1}}{P_t} \quad (1.11)$$

$$= \frac{e_{t+1}}{e_t} \frac{q_{t+1}}{q_t} + \frac{d_{t+1}}{P_t} \quad (1.12)$$

$$= (1 + g_{t+1}^{\text{real}}) \frac{q_{t+1}}{q_t} + y_{t+1}. \quad (1.13)$$

Equivalently, using $d_{t+1} = \kappa e_{t+1}$,

$$R_{t+1} = (1 + g_{t+1}^{\text{real}}) \frac{q_{t+1} + \kappa}{q_t}, \quad (1.14)$$

and hence the exact log-return identity is

$$\log R_{t+1} = \log(1 + g_{t+1}^{\text{real}}) + \Delta \log q_t + \log\left(1 + \frac{\kappa}{q_{t+1}}\right). \quad (1.15)$$

The final term is the contribution of the distribution; omitting it would produce the log gross capital-gain return, not the log total return.

Following [Bogle \(1991\)](#), use $q_{t+1}/q_t = 1 + \Delta q_t/q_t$. The simple return then satisfies the exact identity

$$r_{t+1} = R_{t+1} - 1 \quad (1.16)$$

$$= y_{t+1} + g_{t+1}^{\text{real}} + \frac{\Delta q_t}{q_t} + g_{t+1}^{\text{real}} \frac{\Delta q_t}{q_t}. \quad (1.17)$$

The last term is the interaction between earnings growth and multiple expansion. When earnings growth and the relative change in the multiple are small, this interaction is

second order. We then obtain Bogle's familiar first-order decomposition:

$$\begin{aligned}
r_{t+1} &\approx y_{t+1} + g_{t+1}^{\text{real}} + \frac{\Delta q_t}{q_t} \\
&\approx y_{t+1} + g_{t+1}^{\text{real}} + \Delta \log q_t \\
&\approx \underbrace{y_{t+1}}_{\text{carry}} + \underbrace{g_{t+1}^{\text{real}}}_{\text{earnings news}} \\
&\quad + \underbrace{\frac{\Delta g}{m - g}}_{\text{long-run growth news}} - \underbrace{\frac{\Delta m}{m - g}}_{\text{required-return news}}.
\end{aligned} \tag{1.18}$$

Thus return has three Bogle components: cash-flow yield, realized earnings growth, and multiple expansion. The underbraces label the first two as carry and current earnings news, and split multiple expansion into long-run growth and required-return news. Equivalently, since $\log R_{t+1} = \log(1 + r_{t+1}) \approx r_{t+1}$ for a small simple return, the same three terms approximate the log gross return.

In the deterministic Gordon model, m and g are fixed, so q_t is fixed and multiple expansion contributes nothing to returns. Moreover, $y_{t+1} = d_{t+1}/P_t = m - g$, so

$$r_{t+1} = (m - g) + g = m. \tag{1.19}$$

The required return m is therefore exactly the total return in the deterministic benchmark. In practice, investors continually revise their views about both long-run growth and required returns. These revisions move the price-earnings multiple, sometimes sharply. The next section extends the Gordon formula to capture this channel.

1.2 Stochastic Long-Run Rates

The deterministic benchmark is useful but too rigid: neither long-run growth nor the required return is known with certainty. We therefore adopt a conditional random-parameter interpretation. Conditional on a possible state, g and m are constant over future horizons, which preserves the Gordon geometric-series calculation, and expected earnings satisfy

$$E[e_{t+s} \mid \mathcal{F}_t, g] = e_t(1 + g)^s. \tag{1.20}$$

At date t , however, investors have a distribution over the latent long-run rates m and g . That conditional distribution changes as information arrives. We write $E_t[\cdot]$ for expectation given the information set $\mathcal{F}_t$.

As a reduced-form valuation rule, we assume that the market averages the state-specific Gordon multiples over this parameter uncertainty. The time- t earnings multiple is then

$$q_t = \kappa E_t \left[\frac{1 + g}{m - g} \right]. \tag{1.21}$$

This expectation must be understood as a valuation-weighted expectation. If E_t is instead interpreted as a physical posterior, the rule additionally assumes that, once the state-specific required return m has been assigned, the stochastic discount factor contributes no further covariance with parameter uncertainty. A general no-arbitrage model would price each future cash flow directly with a stochastic discount factor.

1.2.1 The conditional valuation multiple

Define

$$X := m - g. \quad (1.22)$$

We assume $g > -1$ and $X > 0$ with probability one, $E_t[(1+g)/X] < \infty$, and that the conditional distribution of X is sufficiently concentrated away from zero for a local Taylor approximation to be informative. Let

$$\mu_{X,t} := E_t[X], \quad (1.23)$$

$$\sigma_{X,t} := \sqrt{E_t[(X - \mu_{X,t})^2]}, \quad (1.24)$$

$$\rho_{X,t} := \frac{\sigma_{X,t}}{\mu_{X,t}}. \quad (1.25)$$

Thus $\rho_{X,t}$ is the conditional coefficient of variation of the spread. To make the approximation and its error transparent, write

$$u_t := \frac{X - \mu_{X,t}}{\mu_{X,t}}. \quad (1.26)$$

By construction, $E_t[u_t] = 0$ and $E_t[u_t^2] = \rho_{X,t}^2$. The exact identity

$$\frac{1}{1 + u_t} = 1 - u_t + u_t^2 - \frac{u_t^3}{1 + u_t} \quad (1.27)$$

implies

$$E_t \left[\frac{1}{X} \right] = \frac{1}{\mu_{X,t}} \left[1 + \rho_{X,t}^2 - E_t \left(\frac{u_t^3}{1 + u_t} \right) \right]. \quad (1.28)$$

The final expectation is the Taylor remainder. It shows why the coefficient of variation alone does not control the approximation: skewness and probability mass close to $X = 0$ can be important. A sufficient local condition is, for example, $|u_t| \leq c < 1$ almost surely. Alternatively, one can impose a positive parametric distribution for X and evaluate the expectation directly.

These conditions interact with the learning interpretation adopted above. If beliefs about the latent constants m and g are conjugate Gaussian—the natural specification under which $\mu_{m,t}$ and $\mu_{g,t}$ are posterior martingales—then $X = m - g$ is Gaussian, so $P_t(X \leq 0) > 0$, the assumption $X > 0$ almost surely fails, and $E_t[1/X]$ diverges. Tightening the posterior does not repair this: since $P_t(X \leq 0) = \Phi(-1/\rho_{X,t})$ under Gaussian

beliefs, the left-tail mass at $\rho_{X,t} = 0.25$ is only 3.2×10^{-5} , yet the integral is still infinite. Two repairs are available. One may truncate the posterior below at some $X_{\min} > 0$ and carry the truncation through the moments, or one may specify the spread directly as lognormal. The second is analytically convenient: if $X = e^Z$ with $Z \sim N(a_t, s_t^2)$, then $\mu_{X,t} = e^{a_t + s_t^2/2}$ and $1 + \rho_{X,t}^2 = e^{s_t^2}$, so

$$E_t \left[\frac{1}{X} \right] = e^{-a_t + s_t^2/2} = \frac{1 + \rho_{X,t}^2}{\mu_{X,t}}, \quad (1.29)$$

identically in a_t and s_t . The remainder $E_t[u_t^3/(1 + u_t)]$ then vanishes, and the second-order factor $1 + \rho_{X,t}^2$ is exact rather than approximate. The cost is that m and g cannot both remain Gaussian, so the martingale property must be imposed on beliefs about $\log X$ rather than inherited from conjugate updating of the levels. Matching only the first two moments at the baseline calibration used later, the factor $1 + \rho_{X,t}^2$ is exact under a lognormal X , understates $E_t[1/X]$ by 0.4 percent under a gamma X , and by 2.3 percent under a normal X truncated at zero.

We now retain the numerator $1 + g$. Let $\tilde{g}_t := g - \mu_{g,t}$ and define the scaled conditional covariance

$$c_{gX,t} := \frac{\text{Cov}_t(g, X)}{(1 + \mu_{g,t})\mu_{X,t}}. \quad (1.30)$$

Multiplying the exact reciprocal identity above by $1 + \mu_{g,t} + \tilde{g}_t$ and taking expectations gives

$$\frac{q_t}{\kappa} = \frac{1 + \mu_{g,t}}{\mu_{X,t}} (1 + \rho_{X,t}^2) - \frac{\text{Cov}_t(g, X)}{\mu_{X,t}^2} + \mathcal{R}_t^q, \quad (1.31)$$

$$\mathcal{R}_t^q := \frac{1}{\mu_{X,t}} E_t \left[\tilde{g}_t u_t^2 - (1 + \mu_{g,t} + \tilde{g}_t) \frac{u_t^3}{1 + u_t} \right]. \quad (1.32)$$

When the centered growth and spread deviations are locally of order δ and $1 + u_t$ is bounded away from zero, $\mathcal{R}_t^q = O(\delta^3)$. Neglecting this remainder yields the fuller second-order valuation approximation

$$q_t \approx \kappa \frac{1 + \mu_{g,t}}{\mu_{X,t}} (1 + \rho_{X,t}^2 - c_{gX,t}). \quad (1.33)$$

The covariance term matters because states with unusually high growth can also be states with an unusually small discount-growth spread. If this joint term is ignored and $1 + \mu_{g,t} \approx 1$, equation (1.33) reduces to the simpler low-rate expression

$$q_t \approx \frac{\kappa}{\mu_{X,t}} (1 + \rho_{X,t}^2). \quad (1.34)$$

A lower mean spread raises the multiple through the familiar Gordon mechanism. Within the low-rate, spread-only approximation, a higher conditional variance also raises the multiple when the mean is fixed. More generally, because $1/X$ is convex on the positive half-line, its exact expectation rises under a mean-preserving spread. This Jensen

effect is a partial valuation effect, not a statement about how equilibrium risk premia respond to uncertainty.

For compactness, define

$$H_t := 1 + \rho_{X,t}^2 - c_{gX,t}, \quad (1.35)$$

and suppose the second-order approximation satisfies $H_t > 0$. Taking logarithms of equation (1.33) gives

$$\log q_t \approx \log \kappa + \log(1 + \mu_{g,t}) - \log \mu_{X,t} + \log H_t. \quad (1.36)$$

Writing $\mu_{m,t} := E_t[m]$ and $\mu_{g,t} := E_t[g]$, so that $\mu_{X,t} = \mu_{m,t} - \mu_{g,t}$, the change between two consecutive dates can be written without the auxiliary mean-spread notation:

$$\begin{aligned} \Delta \log q_t &\approx \log \left(\frac{1 + \mu_{g,t+1}}{1 + \mu_{g,t}} \right) - \log \left(\frac{\mu_{m,t+1} - \mu_{g,t+1}}{\mu_{m,t} - \mu_{g,t}} \right) + \log \left(\frac{H_{t+1}}{H_t} \right) \\ &= \log \left(1 + \frac{\Delta \mu_{g,t}}{1 + \mu_{g,t}} \right) - \log \left(1 + \frac{\Delta \mu_{m,t} - \Delta \mu_{g,t}}{\mu_{m,t} - \mu_{g,t}} \right) + \log \left(1 + \frac{\Delta H_t}{H_t} \right). \end{aligned} \quad (1.37)$$

This finite-difference expression keeps H_t unexpanded and makes the two rate channels explicit: holding the other rate fixed, an upward growth revision raises the numerator and narrows the mean discount-growth spread, whereas an upward required-return revision widens that spread. A higher H_t raises the multiple through the bundled dispersion-covariance adjustment. The finite form is useful when belief revisions are not especially small. For small relative revisions, linearizing the three logarithms and collecting the growth and required-return terms gives

$$\Delta \log q_t \approx \left(\frac{1}{1 + \mu_{g,t}} + \frac{1}{\mu_{m,t} - \mu_{g,t}} \right) \Delta \mu_{g,t} - \frac{\Delta \mu_{m,t}}{\mu_{m,t} - \mu_{g,t}} + \frac{\Delta H_t}{H_t} \quad (1.38)$$

$$= \frac{1}{\mu_{m,t} - \mu_{g,t}} \left(\frac{1 + \mu_{m,t}}{1 + \mu_{g,t}} \Delta \mu_{g,t} - \Delta \mu_{m,t} \right) + \frac{\Delta H_t}{H_t}. \quad (1.39)$$

Thus the absolute growth-to-required-return loading ratio is $(1 + \mu_{m,t}) / (1 + \mu_{g,t})$, which is close to one when the two rates are small. If the numerator-growth channel is deliberately suppressed, so this ratio is replaced by one, and growth-spread covariance is set to zero, equation (1.39) reduces to

$$\Delta \log q_t \approx \frac{\Delta \mu_{g,t} - \Delta \mu_{m,t}}{\mu_{m,t} - \mu_{g,t}} + \frac{2\rho_{X,t}}{1 + \rho_{X,t}^2} \Delta \rho_{X,t}. \quad (1.40)$$

This compact expression is often adequate because $1/(1 + \mu_{g,t})$ is small relative to the Gordon duration $1/(\mu_{m,t} - \mu_{g,t})$. It should nevertheless be recognized as an additional low-rate and zero-covariance approximation. Under the fixed-latent-parameter interpretation, $\mu_{g,t}$ and $\mu_{m,t}$ are martingales, so their one-period changes are already conditional mean-zero news. Relative dispersion need not be a martingale: ordinary learning can make it decline predictably. If g and m are instead treated as time-varying states, “news” should refer to the innovations in these changes, for example $\Delta \mu_{g,t} - E_t[\Delta \mu_{g,t}]$, rather than to the raw changes themselves.

1.2.2 Earnings news and the return decomposition

At date t , investors forecast next-period earnings growth using $\mu_{g,t}$. Define the realized earnings-growth surprise by

$$\epsilon_{t+1} := g_{t+1}^{\text{real}} - \mu_{g,t}, \quad E_t[\epsilon_{t+1}] = 0. \quad (1.41)$$

The distinction between ϵ_{t+1} and $\Delta\mu_{g,t}$ is important. The former is the surprise in current realized earnings growth; the latter is the revision to expected long-run growth after date- $t+1$ information arrives. A strong earnings realization need not change long-run growth beliefs, while the same signal can move both quantities.

Under constant payout, the conditional expected cash-flow yield and its realization are

$$\bar{y}_t := E_t[y_{t+1}] = \frac{\kappa(1 + \mu_{g,t})}{q_t}, \quad (1.42)$$

$$y_{t+1} = \bar{y}_t + \frac{\kappa}{q_t} \epsilon_{t+1}. \quad (1.43)$$

An earnings surprise therefore affects both earnings growth and the distribution received during the holding period. Its direct return loading is $1 + \kappa/q_t$.

Define $r_{t+1}^{\text{ex}} := r_{t+1} - r_t^f$ and $h_{t+1} := \Delta \log q_t$. The exact arithmetic-return identity implies

$$\begin{aligned} r_{t+1}^{\text{ex}} &= \bar{y}_t + \left(1 + \frac{\kappa}{q_t}\right) \epsilon_{t+1} + (\mu_{g,t} - r_t^f) \\ &\quad + (1 + \mu_{g,t} + \epsilon_{t+1})(e^{h_{t+1}} - 1). \end{aligned}$$

For a local attribution, replace the final term by h_{t+1} and substitute equation (1.39). This requires the multiple change and the products $\mu_{g,t}h_{t+1}$ and $\epsilon_{t+1}h_{t+1}$ to be small. The central stock-level decomposition is then

$$\begin{aligned} r_{t+1}^{\text{ex}} \approx & \underbrace{\bar{y}_t + (\mu_{g,t} - r_t^f)}_{\text{carry}} \\ & + \underbrace{\left(1 + \frac{\kappa}{q_t}\right) \epsilon_{t+1}}_{\text{current earnings news}} \\ & + \underbrace{\left(\frac{1}{1 + \mu_{g,t}} + \frac{1}{\mu_{m,t} - \mu_{g,t}}\right) \Delta\mu_{g,t}}_{\text{long-run growth news}} \\ & - \underbrace{\frac{\Delta\mu_{m,t}}{\mu_{m,t} - \mu_{g,t}}}_{\text{required-return news}} + \underbrace{\frac{\Delta H_t}{H_t}}_{\text{dispersion-covariance news}}. \end{aligned} \quad (1.44)$$

This form collects the long-run growth and required-return revisions and keeps the dispersion-covariance adjustment bundled in H_t . If that last channel is expanded locally,

$$\frac{\Delta H_t}{H_t} \approx \frac{2\rho_{X,t}\Delta\rho_{X,t} - \Delta c_{gX,t}}{H_t}.$$

The first underbrace in equation (1.44) collects conditional carry: the expected cash-flow yield plus expected growth in excess of the risk-free rate. The second is the direct effect of the current earnings surprise. The remaining underbraces are multiple changes caused by revisions to long-run growth, required returns, and H_t . The direct loading $1 + \kappa/q_t$ is not a total empirical earnings-response coefficient, because one earnings signal may move all four state variables simultaneously.

The components are economic attributions, not statistically orthogonal shocks. For large stock-level multiple changes, the exact arithmetic identity above should be used instead of the local approximation. As a consistency check, if beliefs are degenerate, $\epsilon_{t+1} = 0$, and no state variable changes, then $\tilde{y}_t = m - g$ and the equation reduces to $r_{t+1}^{\text{ex}} = m - r_t^f$. The stock-level calibration and empirical importance of the terms are considered next.

1.2.3 Choosing the Gordon discount rate from a multifactor model

The practical answer is to use the stock's long-run factor-implied cost of equity. In each state of the Gordon model, set

$$m_i := r^{f,\text{LR}} + (\boldsymbol{\beta}_i^{\text{LR}})^\top \boldsymbol{\lambda}^{\text{LR}} = r^{f,\text{LR}} + \sum_{k=1}^K \beta_{ik}^{\text{LR}} \lambda_k^{\text{LR}}. \quad (1.45)$$

The superscript LR means the values expected to prevail in the long-run or terminal regime. This distinction is essential: the Gordon model applies m_i to every future cash flow, so a temporary short rate or a temporarily high factor premium is not the appropriate input.

At date t , a baseline plug-in estimate of the conditional mean rate is

$$\hat{\mu}_{m,i,t} := r_t^{f,\text{LR}} + (\hat{\boldsymbol{\beta}}_{i,t}^{\text{LR}})^\top \hat{\boldsymbol{\lambda}}_t^{\text{LR}}. \quad (1.46)$$

The three inputs have distinct empirical roles:

- $r_t^{f,\text{LR}}$ is a normalized long-run risk-free rate in the same currency and nominal or real units as the cash-flow growth rate.
- $\hat{\boldsymbol{\beta}}_{i,t}^{\text{LR}}$ contains the stock's terminal-regime factor exposures. A return regression provides an initial estimate, but noisy stock-level betas should normally be shrunk toward economically relevant industry or market exposures and adjusted for the firm's expected terminal leverage and business mix.

- $\hat{\lambda}_t^{\text{LR}}$ contains sustainable long-run factor premia. For traded excess-return factors these are expected factor returns, not their most recent realizations. For nontraded factors they must be estimated as cross-sectional prices of risk.

The CAPM (Sharpe, 1964) sets $K = 1$. A multifactor implementation may add other priced risks (e.g. Fama and French, 1993). The key practical point is that a time-series regression estimates β_i , not m_i : the factor premia λ and the risk-free rate must be supplied separately. Regression alpha is not added to equation (1.46); it is a pricing error or a return forecast, not compensation for an included priced risk.

Using equation (1.45) requires the following assumptions:

1. The selected factor model prices the stock with zero conditional alpha and includes the systematic risks for which investors require compensation. Idiosyncratic volatility is not added separately unless the model explicitly prices it.
2. The long-run risk-free rate, exposures, and premia are constant, or are sufficiently persistent that one flat rate is a reasonable summary of the rates applied to distant cash flows.
3. All rates use the same period, currency, and nominal or real convention. In particular, nominal growth must be paired with a nominal discount rate.
4. The resulting rate satisfies $m_i > g_i > -1$ by an economically meaningful margin. If it does not, the Gordon perpetuity is not valid; one should revisit the terminal assumptions or use a finite-horizon model rather than mechanically imposing a floor on $m_i - g_i$.

The operational choice among current, long-run, and horizon-specific inputs is:

1. **One-period return analysis.** Use current conditional inputs,

$$m_{i,t}^F := r_t^f + \beta_{i,t}^\top \lambda_t. \quad (1.47)$$

This is the next-period expected return, not automatically a Gordon perpetuity rate.

2. **Gordon or terminal-value analysis.** Use the normalized long-run inputs in equation (1.46). This is the default rule for the model in this chapter. Current inputs may be used only when they are also credible long-run inputs.
3. **Material mean reversion or changing exposures.** Do not force the current one-period rate into a perpetuity. Forecast the horizon-specific required returns and value the cash flows from the resulting rate path:

$$k_{i,t,j} := r_{t,j}^f + \beta_{i,t,j}^\top \lambda_{t,j}, \quad j \geq 1, \quad (1.48)$$

$$P_{i,t}^F := \sum_{s=1}^{\infty} \frac{E_t[d_{i,t+s}]}{\prod_{j=1}^s (1 + k_{i,t,j})}. \quad (1.49)$$

If a single flat rate is still needed after the term-structure valuation, use the constant-rate equivalent. For the Gordon cash-flow path it is

$$m_{i,t}^{\text{CE}} := g_i + \frac{d_{i,t+1}}{P_{i,t}^F} = g_i + \frac{\kappa_i e_{i,t} (1 + g_i)}{P_{i,t}^F}. \quad (1.50)$$

This rate reproduces $P_{i,t}^F$ in a flat-rate Gordon formula. A standard one-period factor regression cannot produce it without forecasts of future risk-free rates, exposures, and premia.

If the long-run factor inputs are uncertain, equation (1.45) should be applied state by state. In particular,

$$\begin{aligned} \mu_{m,i,t} &= E_t[r^{f,\text{LR}}] + E_t[\beta_i^{\text{LR}}]^\top E_t[\lambda^{\text{LR}}] \\ &\quad + \text{tr Cov}_t(\beta_i^{\text{LR}}, \lambda^{\text{LR}}). \end{aligned} \quad (1.51)$$

The plug-in estimate in equation (1.46) ignores the last term. More importantly, valuation should retain the state-specific m_i inside $E_t[(1 + g_i)/(m_i - g_i)]$ rather than replace the nonlinear expectation with a multiple evaluated only at $\mu_{m,i,t}$.

Finally, only revisions to the long-run inputs should enter the Gordon multiple-news term. The exact change in the plug-in estimate and its first-order approximation are

$$\begin{aligned} \Delta \hat{\mu}_{m,i,t} &= \Delta r_t^{f,\text{LR}} + (\Delta \hat{\beta}_{i,t}^{\text{LR}})^\top \hat{\lambda}_t^{\text{LR}} + (\hat{\beta}_{i,t}^{\text{LR}})^\top \Delta \hat{\lambda}_t^{\text{LR}} \\ &\quad + (\Delta \hat{\beta}_{i,t}^{\text{LR}})^\top \Delta \hat{\lambda}_t^{\text{LR}} \\ &\approx \Delta r_t^{f,\text{LR}} + (\Delta \hat{\beta}_{i,t}^{\text{LR}})^\top \hat{\lambda}_t^{\text{LR}} + (\hat{\beta}_{i,t}^{\text{LR}})^\top \Delta \hat{\lambda}_t^{\text{LR}}. \end{aligned} \quad (1.52)$$

Holding growth beliefs and $H_{i,t}$ fixed, its local contribution is

$$\Delta \log q_{i,t}|_m \approx -\frac{\Delta \hat{\mu}_{m,i,t}}{\hat{\mu}_{m,i,t} - \hat{\mu}_{g,i,t}}. \quad (1.53)$$

The next subsection uses a stock-level calibration to illustrate this channel. A temporary policy-rate movement should not be treated as a permanent Gordon-rate change unless it also changes the expected long-run risk-free rate.

The practical rule is therefore: use current factor inputs for one-period return analysis, normalized long-run inputs for a Gordon valuation, and a full rate path when mean reversion or changing exposures is material. This remains a reduced-form use of expected returns as discount rates; a fully structural valuation would instead price each cash flow with a stochastic discount factor.

1.2.4 Empirical magnitudes and relative importance at the stock level

To give equation (1.44) a stock-level scale, consider a mature, positive-earnings dividend payer with $\kappa = 0.40$, $q_t = 21$, $\mu_{g,t} = 5$ percent, $\rho_{X,t} = 0.25$, $c_{gX,t} = 0$, and a long-run discount rate $\mu_{m,t} = 7.125$ percent. The discount rate is the long-run factor-implied cost of equity defined in the preceding subsection, not a next-period return forecast. For example, it is generated by a 3.00 percent long-run risk-free rate, market, size, and value betas $(0.90, 0.25, 0.25)^\top$, and corresponding factor premia $(4.00, 0.80, 1.30)^\top$ percent:

$$\mu_{m,t} = 3.00\% + (0.90)(4.00\%) + (0.25)(0.80\%) + (0.25)(1.30\%) = 7.125\%. \quad (1.54)$$

The resulting discount-growth spread and cash-flow quantities are

$$\mu_{X,t} = \mu_{m,t} - \mu_{g,t} = 2.125 \text{ percent}, \quad \bar{y}_t = \frac{\kappa(1 + \mu_{g,t})}{q_t} = 2.00 \text{ percent}, \quad (1.55)$$

$$\frac{\kappa}{q_t} = \frac{\bar{y}_t}{1 + \mu_{g,t}} \simeq 1.90 \text{ percent}. \quad (1.56)$$

This is an illustrative calibration for a mature payer, not an estimate for the cross-sectional median listed firm. The empirical unit in this subsection is an individual stock-year, not a diversified portfolio, and “representative” requires qualification: payout, growth, volatility, and the mix of news vary sharply with firm size and lifecycle. In particular, the five-percent $\mu_{g,t}$ is a terminal-growth assumption and should not be replaced mechanically by a noisy one- or five-year realized EPS growth rate. Moreover, equation (1.44) is not an orthogonal variance decomposition. Its terms can be correlated, and the empirical studies below use log-linear present-value decompositions rather than exactly the simple-return approximation used here. Their “cash-flow news” is closest to the combination of the earnings surprise and the growth-expectation revision in our equation, while their “expected-return news” is closest to our required-return revision. Campbell-style news revises an entire discounted path, whereas the Gordon terms represent permanent parallel changes in flat long-run parameters. The mapping is therefore qualitative rather than one-to-one (Campbell, 1991).

Two firm-level samples give the relevant empirical scale. In 99,441 U.S. firm-years from 1998–2020, Sloan and Wang (2025) report a mean dividend yield of 1.04 percent, a median of zero, a 75th percentile of 1.45 percent, and a 90th percentile of 3.33 percent. About 37 percent of firms have negative base-year recurring operating EPS. Among surviving firms with positive base-year operating EPS, one-year EPS growth has a median of 8.36 percent and an interquartile range from -25.07 to 33.50 percent; five-year annualized EPS growth has a median of 9.78 percent and an interquartile range from 0.05 to 19.06 percent. These growth statistics are conditional on positive base-year earnings and subsequent survival; their per-share construction adjusts for splits and dividends and assumes that cash dividends and special distributions are reinvested. They are therefore not a plug-in estimate of the continuing-share growth rate in this model without a corresponding payout adjustment. They measure medium-horizon firm heterogeneity, not a sustainable perpetual growth rate. In a separate panel of 36,791 firm-years, Vuolteenaho (2002) report annual excess-log-return volatility of 32.2 percent and market-adjusted

volatility of 28.7 percent. The roughly 30-percent scale is a screened-firm benchmark and is already materially above diversified-index volatility; it should not be read as an unconditional equal-weight average over all listed stocks. The dividend yields above are trailing empirical yields, whereas $\bar{y}_t$ is a forward conditional yield; they identify the stock-level scale, not the same conditional object. The cross-sectional median firm pays no dividend and is not well represented by the constant-positive-payout calibration above.

1. **Cash-flow yield anchors the conditional mean, but there is no positive-yield “typical stock.”** The cross-sectional median dividend yield is zero, so a four-percent index yield is not representative at the stock level. For the mature payer above, the expected one-period cash-flow yield is two percent. In the present per-share model, d_t is the distribution received on one continuing share. If aggregate earnings are E_t^{agg} and shares outstanding are N_t , then

$$e_t = \frac{E_t^{\text{agg}}}{N_t}, \quad \Delta \log e_t = \Delta \log E_t^{\text{agg}} - \Delta \log N_t. \quad (1.57)$$

Repurchases therefore enter through share-count and per-share-growth dynamics. Adding repurchase yield to d_t while retaining EPS growth that already reflects the lower share count would double count the repurchase. A total-payout implementation instead requires aggregate payout net of equity issuance and a market-capitalization formulation.

2. **Bundled cash-flow and discount-rate news are comparable at one year; cash-flow news leads beyond one year.** At the firm level, [Vuolteenaho \(2002\)](#) estimate cash-flow-news and expected-return-news standard deviations of approximately 32 and 22 percent for excess returns. Their variance ratio is therefore

$$\frac{\text{Var}(N_{CF,t})}{\text{Var}(N_{DR,t})} \simeq \left(\frac{0.32}{0.22} \right)^2 \simeq 2.1. \quad (1.58)$$

After removing the market return, the corresponding standard deviations are about 28 and 13 percent, for a variance ratio close to five. These ratios are not variance shares because the two news terms are positively correlated. Using direct analyst cash-flow forecasts rather than backing cash-flow news out as a residual, [Chen et al. \(2013\)](#) report average firm-level covariance-decomposition coefficients of 0.48 for cash-flow news and 0.52 for discount-rate news at one year, 0.63 and 0.37 at two years, and 0.68 and 0.32 at three and four years. Because their construction makes the two components add to cumulative capital-gain return, the coefficients sum to one; they are not component-variance ratios or regression R^2 values. The cash-flow coefficient remains larger at longer horizons, although it declines to 0.62 by year seven. The precise split is therefore method- and horizon-dependent: the two components are approximately tied at one year, while cash-flow news dominates beyond one year.

In the local Gordon equation, the direct response to the contemporaneous earnings surprise is only

$$1 + \frac{K}{q_t} \simeq 1.019. \quad (1.59)$$

The 32-percent empirical cash-flow-news volatility must not be assigned to ϵ_{t+1} alone. It also bundles revisions to expected future cash flows, represented here by

$$\left(\frac{1}{1 + \mu_{g,t}} + \frac{1}{\mu_{X,t}} \right) \Delta \mu_{g,t}. \quad (1.60)$$

3. **Growth and required-return revisions have large and nearly symmetric stock-level Gordon loadings.** At the illustrative stock calibration,

$$\frac{1}{\mu_{X,t}} \simeq 47.06, \quad \frac{1}{1 + \mu_{g,t}} + \frac{1}{\mu_{X,t}} \simeq 48.01. \quad (1.61)$$

Thus a 25-basis-point permanent increase in the flat required return contributes -11.76 percent under the local formula. The finite log-multiple and simple multiple changes from equation (1.37) are -11.12 and -10.53 percent. For permanent increases of 50 and 100 basis points, the corresponding triples—local, finite log, and finite simple multiple—are $(-23.53, -21.13, -19.05)$ percent and $(-47.06, -38.57, -32.00)$ percent. These are deliberately permanent flat-rate shifts; a temporary one-period policy-rate change is not comparable. The exact arithmetic-return contribution of a multiple change is $(1 + g_{t+1}^{\text{real}})(q_{t+1}/q_t - 1)$. The calculation also shows why the former four-percent index-yield calibration understated the duration of a lower-yield individual stock. It is not, however, an empirical median-duration estimate. In 126,870 U.S. firm-years, [Dechow et al. \(2004\)](#) report a median implied equity duration of 15.63 years, with an interquartile range of 13.30–17.36. At their 12-percent maintained discount rate, the median modified duration is about $15.63/1.12 = 13.96$, so a 25-basis-point parallel shift corresponds locally to about -3.5 percent. The difference from -11.76 percent quantifies how strong the flat-growth perpetuity restriction is; the two calculations should not be treated as estimates of the same object.

4. **Spread dispersion and covariance remain empirically unranked at the stock level.** The quantity $\rho_{X,t}$ is a posterior coefficient of variation for the latent long-run spread $m - g$, not an observed volatility index, while $c_{gX,t}$ records a scaled conditional dependence. At the stylized values $\rho_{X,t} = 0.25$ and $c_{gX,t} = 0$, the local dispersion coefficient is

$$\frac{2\rho_{X,t}}{1 + \rho_{X,t}^2} \simeq 0.47. \quad (1.62)$$

A 0.02 increase in $\rho_{X,t}$ contributes about 0.94 percent locally, and a 0.01 increase in $c_{gX,t}$ contributes about -0.94 percent, holding the other state variables fixed. These are sensitivities, not estimates of the distributions of $\Delta \rho_{X,t}$ or $\Delta c_{gX,t}$. [Pástor and Veronesi \(2003\)](#) provide firm-level evidence that uncertainty affects valuation, but do not identify these posterior moments. If an uncertainty shock also raises required risk premia, the negative $-\Delta \mu_{m,t}/(\mu_{m,t} - \mu_{g,t})$ effect can offset or dominate the positive partial-convexity effect.

The defensible stock-level conclusion is therefore horizon-dependent. At one year, bundled cash-flow and growth news is either first or approximately tied with expected-return news, depending on the measurement method; over longer horizons, the direct-forecast evidence ranks cash-flow news first. The posterior-dispersion terms are not yet separately ranked. This is a ranking of empirical variation, not of local coefficients: growth and required-return revisions have almost the same Gordon duration, while their realized volatilities determine their relative contributions. The expected cash-flow yield and expected growth net of the risk-free rate remain firm-specific carry terms and should be calibrated from the stock's own payout, per-share forecasts, and valuation multiple.

Excess-return decomposition summary.

Term	Sign	Economic interpretation	Relative impact at the stock level	Connection to Bogle
$\bar{y}_t$	+	Conditional expected cash-flow yield.	Carry, not news: trailing dividend yield is zero at the median, 1.04 percent at the mean, and 2 percent for the illustrative mature payer.	CF yield (conditional expected component).
$\left(1 + \frac{\kappa}{q_t}\right) \epsilon_{t+1}$	Same sign as ϵ_{t+1}	Direct earnings forecast error affecting both the capital gain and the contemporaneous distribution.	Material but not separately identified: the loading is 1.019, while the 28–32 percent empirical cash-flow-news volatility also contains growth revisions.	CF yield + fundamental growth: $(\kappa/q_t)\epsilon_{t+1}$ is the yield surprise and ϵ_{t+1} is realized growth news.
$\mu_{g,t} - r_t^f$	+ when > 0	Expected earnings growth net of the risk-free rate: the baseline conditional excess-return component.	Stock-specific conditional carry, not a news shock; the cited variance decompositions do not rank this term.	Fundamental growth; $-r_t^f$ is the excess-return adjustment, not a Bogle term.
$-\frac{\Delta\mu_{m,t}}{\mu_{m,t} - \mu_{g,t}}$	– when $\Delta\mu_{m,t} > 0$	A higher flat required return compresses the valuation multiple.	Large valuation sensitivity: 25 basis points gives –11.76 percent in the Gordon calibration versus about –3.5 percent at the empirical median duration. Discount-rate news is about half of one-year firm-price variation.	Multiple expansion (discount-rate-induced; compression when the rate rises).
$\frac{\Delta\mu_{g,t}}{1 + \mu_{g,t}} + \frac{\Delta\mu_{g,t}}{\mu_{m,t} - \mu_{g,t}}$	Same sign as $\Delta\mu_{g,t}$	An upward growth revision raises the Gordon numerator and the value of the perpetuity.	Large and almost symmetric with the required-return revision (48.01 versus 47.06). Bundled cash-flow news dominates beyond one year.	Multiple expansion (expected-growth-induced): the revision changes q_t , not current realized growth.
$\frac{\Delta H_t}{H_t} \approx \frac{2\rho_{X,t}\Delta\rho_{X,t}}{H_t} - \frac{\Delta c_{gX,t}}{H_t}$	Depends on the state changes	Higher relative dispersion raises the multiple; higher growth-spread covariance lowers it, holding other states fixed.	Empirically unranked: stylized state changes give about ± 0.94 percent, and an accompanying risk-premium response may dominate.	Multiple expansion (uncertainty-induced).

Comparison note: the final column maps each row to Bogle's three components in equation (1.18). The other entries mix conditional carry, empirical news variation, and local Gordon sensitivities; because the terms are correlated, they are not additive variance shares.

1.3 ★ A Different Model: Persistent Means with Lognormal Rate Fluctuations

The preceding model treats the uncertain long-run rates g and m as latent constants. A simple extension separates this persistent uncertainty from period-by-period variation. Draw the latent centers (μ_g, μ_m) once and hold them fixed over all future horizons. Conditional on those centers, draw new growth and discount-rate shocks each period. The fixed parameters τ_g and τ_m control the magnitudes of the transitory shocks. Here the undated μ_g and μ_m are path-specific latent rates; their date- t posterior estimates are introduced below. Conditional independence restores a geometric present-value series.

1.3.1 A mean-preserving lognormal specification

Write the future gross growth and gross discount rates as

$$\begin{aligned} G_{t+s} &:= 1 + g_{t+s} = (1 + \mu_g) \exp \left(-\frac{\tau_g^2}{2} + \tau_g \eta_{t+s}^g \right), \\ M_{t+s} &:= 1 + m_{t+s} = (1 + \mu_m) \exp \left(-\frac{\tau_m^2}{2} + \tau_m \eta_{t+s}^m \right), \quad s \geq 1, \end{aligned} \quad (1.63)$$

where $\mu_g, \mu_m > -1$, $\tau_g, \tau_m \geq 0$, and

$$\eta_{t+s}^g, \eta_{t+s}^m \stackrel{\text{iid}}{\sim} N(0, 1). \quad (1.64)$$

The innovations are mutually independent across rates and dates and are independent of the persistent centers. The $-\tau_j^2/2$ terms make the specification mean-preserving:

$$E[G_{t+s} \mid \mu_g, \mu_m] = 1 + \mu_g, \quad E[M_{t+s} \mid \mu_g, \mu_m] = 1 + \mu_m. \quad (1.65)$$

Thus μ_g and μ_m retain their interpretation as conditional arithmetic mean rates. The simple rates can be negative, but their gross counterparts are always positive.

Parameterization note. If $1 + \mu_j$ is instead intended to be the conditional median, omit the compensator $-\tau_j^2/2$. Then replace the wedge ω_τ below by $\omega_\tau^{\text{med}} = (\tau_g^2 + \tau_m^2)/2$. The distinction is second order in the log volatilities.

1.3.2 The conditional multiple is again a Gordon fraction

Conditional on the persistent centers, define the expected one-period growth-to-discount ratio

$$\begin{aligned} A(\mu_g, \mu_m) &:= E \left[\frac{G_{t+s}}{M_{t+s}} \mid \mu_g, \mu_m \right] \\ &= \frac{1 + \mu_g}{1 + \mu_m} \exp(\omega_\tau), \quad \omega_\tau := \tau_m^2. \end{aligned} \quad (1.66)$$

The term ω_τ is the lognormal discounting wedge. Because the shock pairs are independent across dates,

$$E \left[\prod_{j=1}^s \frac{G_{t+j}}{M_{t+j}} \middle| \mu_g, \mu_m \right] = A(\mu_g, \mu_m)^s. \quad (1.67)$$

The state-specific earnings multiple is therefore

$$\begin{aligned} q(\mu_g, \mu_m) &= \kappa \sum_{s=1}^{\infty} A(\mu_g, \mu_m)^s \\ &= \kappa \frac{A(\mu_g, \mu_m)}{1 - A(\mu_g, \mu_m)}. \end{aligned} \quad (1.68)$$

This formula is exact under equation (1.63); no small-volatility approximation has yet been used.

It is convenient to work with the log centers

$$\theta_g := \log(1 + \mu_g), \quad \theta_m := \log(1 + \mu_m) \quad (1.69)$$

and the effective log discount-growth gap

$$\delta := \theta_m - \theta_g - \omega_\tau. \quad (1.70)$$

Then $A = e^{-\delta}$ and

$$q(\mu_g, \mu_m) = \frac{\kappa}{e^\delta - 1}, \quad q(\mu_g, \mu_m) < \infty \iff \delta > 0. \quad (1.71)$$

The ordinary Gordon formula is recovered when $\tau_g = \tau_m = 0$. Transitory lognormal risk does not destroy the closed form; it only replaces the deterministic log gap $\theta_m - \theta_g$ by the effective gap δ .

This valuation uses M_{t+s}^{-1} as a positive, reduced-form one-period discount factor. It does not by itself imply that μ_m is the asset's conditional expected return. Giving m_{t+s} both interpretations requires a stochastic-discount-factor model that enforces the corresponding return restriction.

The small-volatility regime. Suppose $\mu_m > \mu_g$ and define the zero-volatility multiple by

$$q^{(0)} := \kappa \frac{1 + \mu_g}{\mu_m - \mu_g}. \quad (1.72)$$

If $\tau_\star := \max\{\tau_g, \tau_m\}$ is small and the rates remain away from the convergence boundary, then $\omega_\tau = O(\tau_\star^2)$ and

$$q(\mu_g, \mu_m) = q^{(0)} \left[1 + \frac{1 + \mu_m}{\mu_m - \mu_g} \omega_\tau \right] + O(\tau_\star^4). \quad (1.73)$$

Thus the first volatility correction is second order. When the effective log gap is small, equation (1.71) also gives

$$q(\mu_g, \mu_m) \approx \frac{\kappa}{\log((1 + \mu_m)/(1 + \mu_g)) - \omega_\tau}. \quad (1.74)$$

If quadratic terms in the rates are negligible as well, the denominator can be simplified further to $\mu_m - \mu_g - \omega_\tau$. Either form is useful only when the effective gap is positive and not close to zero.

Discount-rate volatility raises the multiple through the convexity of the inverse discount factor. Mean-preserving growth volatility τ_g affects realized earnings risk but not valuation. With contemporaneous innovation correlation ϱ , the only change is $\omega_\tau = \tau_m^2 - \varrho\tau_g\tau_m$. Near the convergence boundary the coefficient $(1 + \mu_m)/(\mu_m - \mu_g)$ is large, so even small volatilities can have a material valuation effect and the local expansion can fail.

1.3.3 Persistent center uncertainty and multiple news

The latent centers are random at date t but fixed along each future path. Let the time- t posterior distribution of (θ_g, θ_m) represent investors' uncertainty about them. Here E_t has the same valuation-weighted interpretation as in the preceding stochastic-rate model. Averaging the state-specific multiple gives the exact hierarchical valuation rule

$$q_t = \kappa E_t[f(\delta)], \quad f(\delta) := \frac{1}{e^\delta - 1}. \quad (1.75)$$

The expectation remains outside the Gordon fraction because the same latent centers apply at every horizon. Redrawing the centers independently each period would be a different model.

For a finite fundamental value, the posterior must put probability one on $\delta > 0$ and the expectation in equation (1.75) must be finite. A convenient sufficient restriction is $\delta \geq \delta_{\min} > 0$ almost surely. An unrestricted Gaussian posterior for $\theta_m - \theta_g$ is therefore not admissible: it assigns positive probability to a nonpositive effective gap. This is the same boundary problem as in the original random-parameter model, now expressed in a single scalar.

When the posterior is concentrated away from zero, define

$$\bar{\delta}_t := E_t[\delta], \quad v_{\delta,t} := \text{Var}_t(\delta). \quad (1.76)$$

Because ω_τ is fixed,

$$\begin{aligned} \bar{\delta}_t &= E_t[\theta_m] - E_t[\theta_g] - \omega_\tau, \\ v_{\delta,t} &= \text{Var}_t(\theta_m) + \text{Var}_t(\theta_g) - 2 \text{Cov}_t(\theta_g, \theta_m). \end{aligned} \quad (1.77)$$

A second-order Taylor expansion gives

$$q_t \approx \kappa \left[\frac{1}{e^{\bar{\delta}_t} - 1} + \frac{1}{2} \frac{e^{\bar{\delta}_t} (e^{\bar{\delta}_t} + 1)}{(e^{\bar{\delta}_t} - 1)^3} v_{\delta,t} \right]. \quad (1.78)$$

The posterior-variance term is positive because the Gordon multiple is convex in the effective gap. For a small positive $\bar{\delta}_t$, the dominant terms are

$$q_t \approx \kappa \left(\frac{1}{\bar{\delta}_t} + \frac{v_{\delta,t}}{\bar{\delta}_t^3} \right), \quad (1.79)$$

provided $\sqrt{v_{\delta,t}} \ll \bar{\delta}_t$. This formula separates the two uncertainties cleanly: the fixed transitory volatilities shift $\bar{\delta}_t$ through ω_τ , while posterior uncertainty about the persistent centers contributes through $v_{\delta,t}$. Controlling the Taylor remainder also requires bounded standardized higher moments, or an explicit posterior-support restriction away from zero.

Learning and multiple news. For known centers and fixed κ , equation (1.71) has the exact first differential

$$d \log q = \mathcal{D}(\delta) (d\theta_g - d\theta_m + d\omega_\tau), \quad \mathcal{D}(\delta) := \frac{1}{1 - e^{-\delta}}. \quad (1.80)$$

The loading $\mathcal{D}(\delta)$ is a log-gap version of Gordon duration and satisfies $\mathcal{D}(\delta) \approx 1/\delta$ for a small positive gap. Because τ_g and τ_m are fixed in the present model, $d\omega_\tau = 0$ in time-series return attribution. The transitory volatilities affect the level and duration of the multiple, but they are not themselves news.

Let

$$\bar{\theta}_{g,t} := E_t[\theta_g], \quad \bar{\theta}_{m,t} := E_t[\theta_m], \quad (1.81)$$

and define

$$\mathcal{D}_t := \mathcal{D}(\bar{\delta}_t), \quad \mathcal{J}_t := \frac{1}{2} \frac{e^{\bar{\delta}_t} (e^{\bar{\delta}_t} + 1)}{(e^{\bar{\delta}_t} - 1)^2}. \quad (1.82)$$

These are the leading delta-method loadings. Applying the same local approximation to equation (1.78) yields

$$\Delta \log q_t \approx \mathcal{D}_t (\Delta \bar{\theta}_{g,t} - \Delta \bar{\theta}_{m,t}) + \mathcal{J}_t \Delta v_{\delta,t}. \quad (1.83)$$

Terms of order $v_{\delta,t} \Delta \bar{\delta}_t$ and products of belief revisions are omitted; the exact duration of the posterior-averaged multiple is generally not $\mathcal{D}(E_t[\delta])$. An upward revision to the persistent growth center expands the multiple; an upward revision to the persistent discount-rate center compresses it. A rise in posterior uncertainty raises the multiple through convexity. Under a correctly specified fixed-parameter learning model, posterior variance declines in expectation as information accumulates.

1.3.4 Return attribution

No new return identity is required. With $\ell_{t+1} := \log R_{t+1}$,

$$\ell_{t+1} = \log G_{t+1} + \Delta \log q_t + \log \left(1 + \frac{\kappa}{q_{t+1}} \right). \quad (1.84)$$

Define

$$c_t := \log \left(1 + \frac{\kappa}{q_t} \right), \quad \alpha_t := \frac{q_t}{q_t + \kappa}, \quad (1.85)$$

and the direct log-earnings news

$$\begin{aligned} u_{t+1}^g &:= \log G_{t+1} - E_t[\log G_{t+1}] \\ &= (\theta_g - \bar{\theta}_{g,t}) + \tau_g \eta_{t+1}^g. \end{aligned} \quad (1.86)$$

A first-order expansion of the payout term, followed by substitution of equation (1.83), gives the compact attribution

$$\begin{aligned} \ell_{t+1} \approx & \underbrace{\bar{\theta}_{g,t} - \frac{\tau_g^2}{2} + c_t}_{\text{growth and payout carry}} + \underbrace{u_{t+1}^g}_{\text{direct earnings news}} \\ & + \underbrace{\alpha_t \mathcal{D}_t \Delta \bar{\theta}_{g,t}}_{\text{persistent-growth revision}} - \underbrace{\alpha_t \mathcal{D}_t \Delta \bar{\theta}_{m,t}}_{\text{persistent discount-rate revision}} \\ & + \underbrace{\alpha_t \mathcal{F}_t \Delta v_{\delta,t}}_{\text{posterior-uncertainty revision}}. \end{aligned} \quad (1.87)$$

This display is local. It omits the quadratic payout-expansion remainder, the mixed term $v_{\delta,t} \Delta \bar{\delta}_t$, and products of belief revisions. The terms need not be orthogonal. In particular, the same observation that produces u_{t+1}^g can update the posterior center and variance. If the centers are known, then $u_{t+1}^g = \tau_g \eta_{t+1}^g$ and all posterior-revision terms vanish. A transitory discount-rate shock has no direct cash-flow payoff; it affects the ending multiple only to the extent that observing it changes beliefs about the persistent discount-rate center. In short, the random centers govern long-run valuation and learning, whereas the fixed log volatilities govern recurring short-run dispersion.

1.4 Discussion

Relation to the Literature and Contribution

The return accounting used here has close antecedents. [Bogle \(1991\)](#) separates long-horizon equity returns into an investment component—the initial dividend yield plus earnings growth—and a speculative component arising from changes in the price-earnings multiple. [Ferreira and Santa-Clara \(2011\)](#) use the corresponding dividend-price,

earnings-growth, and price-earnings-growth components in their sum-of-the-parts forecasting method. The contribution here is therefore not the three-part accounting identity itself. We derive it from the one-period total-return definition, retain the exact interaction before linearizing, and give changes in the earnings multiple an explicit Gordon-model interpretation.

A complementary literature starts from present-value restrictions. [Campbell and Shiller \(1988a,b\)](#) relate valuation ratios, earnings, and expected future dividends to discount factors, while [Campbell \(1991\)](#) expresses unexpected returns as cash-flow news minus expected-return news. [Chen et al. \(2013\)](#) emphasize that cash-flow news contains both current cash-flow innovations and revisions to expected future cash flows. [Collins and Kothari \(1989\)](#) show that empirical earnings-response coefficients depend on earnings persistence and growth as well as discount rates and risk. The clean-surplus framework of [Ohlson \(1995\)](#) provides a complementary link among earnings, book value, and dividends. [van Binsbergen and Koijen \(2010\)](#) demonstrate how latent expected returns and dividend growth can be inferred jointly in a state-space model subject to a present-value restriction; such dynamics are a natural extension when the latent centers evolve rather than remain fixed. [Böni and Zimmermann \(2021\)](#) use the Gordon model empirically to separate changes in implied long-run growth from changes in discount rates. Richer models of persistent long-run growth risk can also affect equilibrium risk premia and valuation ([Bansal and Yaron, 2004](#)), while [Cochrane \(2011\)](#) reviews the broader evidence on time-varying discount rates. These precedents make clear that the present growth and discount-rate components are economic attributions, not separately identified observables without additional data and filtering assumptions.

The valuation effect of parameter uncertainty is closely related to [Pástor and Veronesi \(2003, 2006\)](#). Under an explicit stochastic-discount-factor construction, [Pástor and Veronesi \(2006\)](#) obtain an expected reciprocal Gordon multiple when long-run growth is uncertain, so Jensen’s inequality raises valuation when the relevant spread is small. The distinctive feature of the present exposition is the combination of this convexity mechanism with Bogle return accounting: uncertainty is assigned to the entire spread $X = m - g$, joint growth-spread dependence is retained locally, and multiple changes are combined with the contemporaneous earnings innovation and posterior-mean revisions. Equation (1.44) should accordingly be viewed as a compact local Gordon attribution, rather than as a new structural or statistically orthogonal return decomposition.

The hierarchical lognormal extension separates posterior uncertainty about fixed long-run centers from transitory period-by-period dispersion. Conditional independence preserves an exact Gordon fraction with an effective log gap, while uncertainty about the fixed centers remains outside that nonlinear fraction. Fixed log volatilities therefore change the level and duration of the multiple but generate no separate multiple-news term.

Key Insights

- The exact simple return is separated from its local approximation. The latter attributes excess return to expected cash-flow yield, the direct earnings forecast er-

ror, expected growth relative to the risk-free rate, revisions to long-run growth and discount rates, and changes in relative spread dispersion and growth-spread dependence.

- Under constant payout, an earnings surprise affects both the ending earnings level and the distribution paid over the holding period. Its direct loading $1 + \kappa/q_t$ is not the total empirical response when the same signal also changes posterior beliefs.
- The decomposition makes clear that realized returns can be high even when fundamentals do not improve, if expected discount rates fall (valuation expansion).
- For a Gordon valuation, the discount rate should be the long-run factor-implied cost of equity. Current one-period factor inputs are appropriate only if they are also credible long-run inputs; otherwise one should use normalized terminal inputs or a horizon-specific discount-rate path.
- Holding the mean spread and growth-spread dependence fixed, greater spread dispersion raises the current multiple through convexity but lowers its subsequent cash-flow yield. If uncertainty also raises required factor premia, the mean-spread effect works in the opposite direction.
- With independent, mean-preserving lognormal shocks, discount-rate volatility raises valuation through inverse-discount convexity, whereas growth volatility affects realized earnings risk but not the conditional multiple. Fixed log volatilities alter valuation levels and loadings rather than creating news by themselves.
- A key empirical implication is that return predictability should come from cash-flow yields and state variables linked to expected discount rates and expected growth, rather than from contemporaneous earnings surprises alone.

Open Problems

- **Identification and state-space estimation:** price and earnings data alone identify only combinations of expected growth, required returns, and spread uncertainty. One possible dynamic empirical extension sets $\mu_{X,t} = \exp(\ell_{X,t})$, $\rho_{X,t} = \exp(h_t)$, and $\mu_{m,t} = \mu_{g,t} + \exp(\ell_{X,t})$, and gives the latent state $\mathbf{s}_t = (\mu_{g,t}, \ell_{X,t}, h_t)^\top$ a transition law. With $c_{gX,t} = 0$ as an initial identifying restriction, natural measurement equations include

$$g_{t+1}^{\text{real}} = \mu_{g,t} + \varepsilon_{t+1}^e, \quad (1.88)$$

$$\log q_t \approx \log \kappa + \log(1 + \mu_{g,t}) - \ell_{X,t} + \log(1 + \exp(2h_t)) + \varepsilon_t^q. \quad (1.89)$$

Analyst long-run growth forecasts can provide a noisy measurement of $\mu_{g,t}$, while survey or factor-model required-return estimates can provide a noisy measurement of $\mu_{m,t}$. A model-specific uncertainty proxy is needed to identify h_t , and separate information is required before $c_{gX,t}$ can be estimated. The long-run factor estimate in equation (1.46) should enter as a noisy measurement of $\mu_{m,t}$, not as an accounting identity.

- **Dynamics of uncertainty:** modeling and forecasting $\rho_{X,t}$ and $c_{gX,t}$ in the presence of learning, structural breaks, and macro stress remains open. A pure fixed-parameter learning model predicts information resolution; persistent regime variation requires time-varying states.
- **Earnings versus cash flows:** empirical work should choose either per-share dividends with explicit share-count dynamics, or aggregate payout with net equity issuance. The two formulations should not be mixed.
- **Risk-premium microfoundations:** connecting reduced-form discount-rate beliefs and the valuation-weighted parameter distribution to an explicit stochastic discount factor and a term structure of expected returns remains open.
- **Approximation validation:** simulations or an empirical appendix should compare the exact nonlinear return and posterior expectation with the local return and second-order valuation approximations over the intended data frequency.
- **Cross-sectional extension:** extending the decomposition to multi-asset settings with heterogeneous firms, sectoral exposures, and correlated earnings/discount-rate shocks.

2 Return Decomposition at a Glance

Figure 1 groups the terms in the local excess-return attribution in equation (1.44). The four branches are additive contributions to the approximation, but they need not be statistically orthogonal. The second-order branch refers to posterior dispersion in the latent long-run spread, not to the fixed log volatilities in Section 1.3.

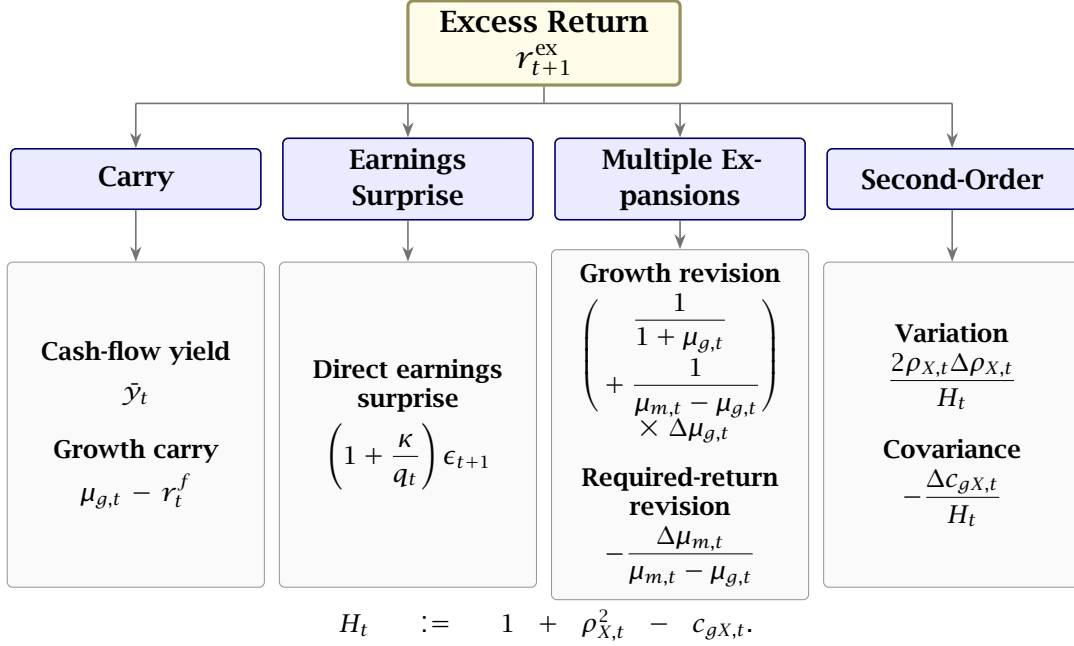


Figure 1: The local excess-return decomposition.

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